A Ten-Year Savings and Investment Plan for a New College Graduate

70-492 Investments

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Prepared for:	New College Graduate Client
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Date:	April 2026
Horizon:	10 Years (May 2026 – May 2036)
Goal:	$150,000 (real, 2026 dollars)
Recommended Allocation:	60% IVV · 13% USMV · 27% SHV

Investment Recommendation at a Glance

Asset	Weight	Role
IVV (iShares S&P 500 ETF)	60.4%	Equity risk premium
USMV (iShares Min-Vol ETF)	12.6%	Low-volatility factor tilt
SHV (iShares Short Treasury)	27.0%	Risk-free / emergency buffer
Portfolio Sharpe Ratio	0.87	
Goal Attainment Probability	61.6%	
Median Terminal Wealth (real)	$157,068	
5th-Percentile Floor (real)	$122,760	

All wealth figures are expressed in 2026 real dollars, deflated at 2.5% per annum. Historical returns cover January 2005 – December 2025 (251 months, Yahoo Finance). Simulation uses 5,000 block-bootstrap paths with 12-month blocks.

Abstract

This report presents a complete, quantitatively-grounded savings and investment plan for a 22-year-old college graduate entering the workforce in May 2026. The client earns $65,000 gross in the first year, carries $30,000 in federal student loans, and seeks to accumulate $150,000 in real (2026) dollars within ten years for a residential property down payment.

After projecting year-by-year net savings capacity ($6,093 rising to $16,914), we construct a five-asset universe of exchange-traded funds (IVV, AGG, SHV, QUAL, USMV) and apply mean-variance optimization under no-short-sales constraints to identify the tangency portfolio. Positioning the client on the Capital Allocation Line to match a 20% maximum drawdown tolerance yields a final allocation of **60% IVV / 13% USMV / 27% SHV**. A 5,000-path block-bootstrap simulation finds a goal-attainment probability of **61.6%**, a median terminal real wealth of **$157,068**, and a 5th-percentile floor of **$122,760**—roughly $32,000 above the all-equity alternative at nearly identical expected upside. Stress tests, Fama-French five-factor regressions, and estimation-window sensitivity checks corroborate the recommendation's robustness.

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1. Introduction

Personal financial planning sits at the intersection of household economics and asset-pricing theory. For a new college graduate, the challenge is particularly acute: income is modest and uncertain, liabilities (student debt) are front-loaded, and the investment horizon is long enough to warrant meaningful equity exposure yet short enough that a severe bear market near the goal date poses real danger.

This report develops a coherent, end-to-end plan that answers four practitioner-relevant questions:

1. **How much should the client save each year?**
   We project deterministic net savings from income, taxes, living expenses, and loan amortization over ten years.

2. **In which securities should the savings be invested?**
   We apply mean-variance theory (Markowitz, 1952) to a universe of five low-cost ETFs, identifying the tangency portfolio with and without factor-tilted funds.

3. **How much risk is appropriate?**
   We use the Capital Allocation Line (Tobin, 1958) to translate the client's stated maximum-drawdown tolerance into a precise split between risky and risk-free assets.

4. **Will the client reach the goal, and what can go wrong?**
   A block-bootstrap Monte Carlo simulation quantifies the goal-attainment probability and a battery of stress tests identifies the most damaging adverse scenarios.

Section 2 establishes the client profile and assumptions. Section 3 presents the savings schedule. Section 4 characterizes the investment universe. Section 5 derives the optimal portfolio. Section 6 presents simulation results and strategy comparisons. Section 7 covers risk analysis and stress testing. Section 8 interprets the portfolio through a factor lens. Section 9 assesses robustness to estimation-window choice. Section 10 synthesizes the recommendation.

2. Client Profile and Assumptions

2.1. Demographics and Financial Situation

The client is a 22-year-old college graduate commencing full-time employment in May 2026 in an urban U.S. metro area. Table 1 summarizes the baseline assumptions.

Table 1: Client baseline assumptions

Attribute	Value	Source / Rationale
Age at plan start	22 years	Graduating cohort
Starting gross salary	$65,000	BLS median, B.S. economics / business
Salary growth (Yr 1–4)	5.0%/yr nominal	Early-career wage-growth literature
Salary growth (Yr 5–10)	3.0%/yr nominal	Post-ramp-up slowdown
Effective tax rate	20% on gross	2026 bracket + standard deduction
Monthly living expenses	$3,500 (Yr 1)	Urban, rent-heavy cost of living
Expense growth	2.5%/yr	Matches baseline CPI assumption
Student loan balance	$30,000 at 5.5%	Average federal direct loan
Loan repayment term	10 years (standard)	Fixed monthly payment: $390.69
Baseline inflation	2.5%/yr	Long-run Federal Reserve target
Emergency fund target	6 months expenses	Financial-planning convention

2.2. Investment Goal

The client seeks to accumulate **$150,000 in real (2026) dollars** by May 2036 for a residential property down payment. Expressing the goal in real terms is essential: a nominal target fails to account for the erosion of purchasing power over a decade of inflation. At 2.5% annual inflation, $150,000 real in 2036 corresponds to approximately $192,000 nominal.

2.3. Risk Tolerance

The client's stated risk tolerance is a willingness to absorb a **maximum portfolio drawdown of 20%** without abandoning the investment plan. This single, concrete preference is sufficient to determine the client's position along the Capital Allocation Line (Section 5.3).

3. Savings Capacity and Cash Flow Analysis

3.1. Net Savings Projection

Annual investable savings are computed as:

$$S_t = \text{Salary}_t\,(1-\tau) - E_t - L \tag{1}$$

where $\tau = 0.20$ is the effective tax rate, E_t denotes annual living expenses (growing at inflation), and $L = \$4{,}688$ is the fixed annual loan payment under the standard 10-year amortization schedule.

Table 2: Year-by-year savings schedule (nominal dollars)

Year	Salary	Taxes	Expenses	Loan	Savings	Cumulative
1 (2026)	$65,000	$13,000	$42,000	$4,688	$6,093	$6,093
2 (2027)	$68,250	$13,650	$43,050	$4,688	$7,643	$13,736
3 (2028)	$71,663	$14,333	$44,126	$4,688	$9,297	$23,033
4 (2029)	$75,246	$15,049	$45,229	$4,688	$11,060	$34,093
5 (2030)	$79,008	$15,802	$46,360	$4,688	$12,939	$47,032
6 (2031)	$81,378	$16,276	$47,519	$4,688	$13,676	$60,709
7 (2032)	$83,819	$16,764	$48,707	$4,688	$14,442	$75,150
8 (2033)	$86,334	$17,267	$49,925	$4,688	$15,236	$90,386
9 (2034)	$88,924	$17,785	$51,173	$4,688	$16,059	$106,445
10 (2035)	$91,592	$18,318	$52,452	$4,688	$16,914	$123,359

3.2. The Investment Imperative

Total undiscounted contributions over ten years amount to $123,359— $26,641 short of the $150,000 real goal. After adjusting for inflation, the pure-savings path falls even further short. **Investing is not optional; it is a necessary condition for goal attainment.**

Savings capacity is back-loaded: Year 1 contributions ($6,093) are less than 40% of Year 10 contributions ($16,914), reflecting rising salary against fixed loan payments and relatively stable inflation-adjusted expenses. This back-loading has an important implication for risk allocation—the portfolio is small early on (so large drawdowns are manageable in absolute terms) but accumulates quickly in the final years (so risk-reduction near the goal date becomes valuable).

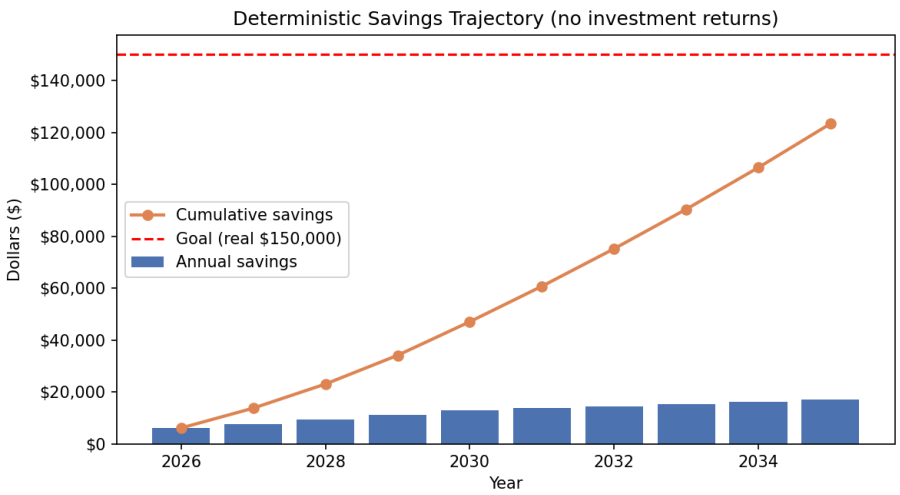


Figure 1: Deterministic savings trajectory: annual contributions (bars) and cumulative savings (orange line) relative to the $150,000 real goal (red dashed). Without investment returns the client falls approximately $27,000 short.

4. Investment Universe and Historical Analysis

4.1. Asset Universe

We employ a two-layer asset universe of exchange-traded funds, all with low expense ratios and sufficient liquidity for a retail investor.

Table 3: Layer A asset universe (IVV, AGG, SHV)

Ticker	Name	Role
IVV	iShares Core S&P 500 ETF	Equity market; captures the U.S. equity risk premium
AGG	iShares Core U.S. Aggregate Bond ETF	Investment-grade bond diversifier
SHV	iShares Short Treasury ETF	Risk-free / cash proxy; emergency fund vehicle

Layer A — Core Three Assets.

Table 4: Layer B asset universe (adds QUAL and USMV)

Ticker	Name	Role	Inception
QUAL	iShares MSCI USA Quality Factor ETF	Profitability tilt	July 2013
USMV	iShares MSCI USA Min Vol Factor ETF	Low-volatility tilt	Oct. 2011

Layer B — Adds Two Factor ETFs. The quality and minimum-volatility factors are well-documented in the asset-pricing literature (Novy-Marx, 2013; Baker, Bradley, and Wurgler, 2011). USMV specifically is motivated by Ang (2014), who documents that low-beta stocks have historically earned higher risk-adjusted returns than the CAPM predicts.

4.2. Historical Return Statistics

Monthly total returns span January 2005 through December 2025 (251 months) for the Layer A universe, and the QUAL/USMV overlap period (January 2013 – December 2025, 145 months) for Layer B analysis. Statistics are annualized as $\hat{\mu}_{\text{ann}} = 12\,\hat{\mu}_m$ and $\hat{\sigma}_{\text{ann}} = \sqrt{12}\,\hat{\sigma}_m$.

Table 5: Annualized historical return statistics (full sample)

Asset	Ann. Return	Ann. Vol	Sharpe (vs SHV)	Max Ann. Drawdown
IVV	13.6%	15.2%	0.80	−51% (2008–09)
AGG	3.0%	3.8%	0.81	−14% (2022)
SHV	1.5%	0.3%	—	≈ 0%
QUAL	13.2%	14.7%	0.79	−38% (2020)
USMV	11.1%	11.1%	**0.86**	−30% (2020)

USMV achieves the highest individual Sharpe ratio (0.86) at roughly 27% lower volatility than IVV, consistent with the low-volatility anomaly. QUAL's return and Sharpe ratio are nearly identical to IVV's, which anticipates the high cross-asset correlation documented below.

4.3. Correlation Structure

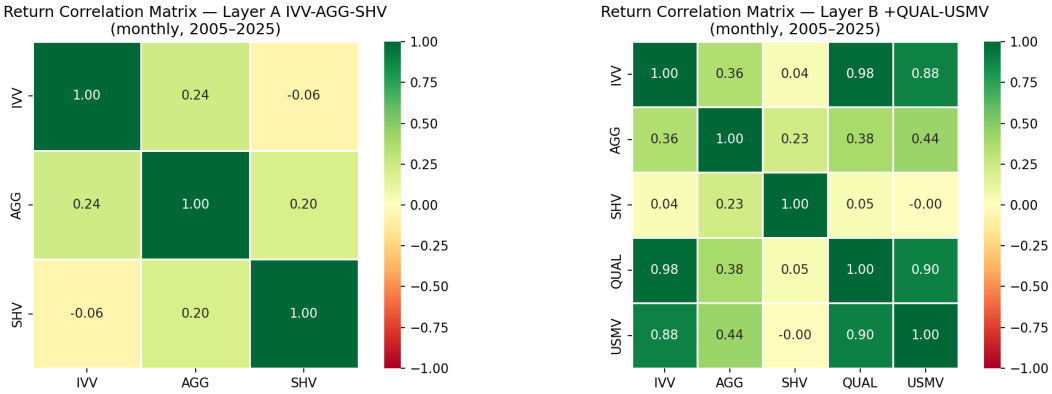
Table 6: Pairwise return correlations — Layer A (2005–2025)

	IVV	AGG	SHV
IVV	1.00	0.02	−0.01
AGG	0.02	1.00	0.24
SHV	−0.01	0.24	1.00

Table 7: Selected pairwise return correlations — Layer B (2013–2025)

	IVV	QUAL	USMV
IVV	1.00	0.98	0.88
QUAL	0.98	1.00	0.88
USMV	0.88	0.88	1.00

The near-zero IVV–AGG correlation ($\rho = 0.02$) in Table 6 confirms the classical equity-bond diversification benefit. Conversely, QUAL's $\rho = 0.98$ with IVV (Table 7) makes it a near-redundant asset; the optimizer will not find an allocation that benefits from including it alongside IVV. USMV's more modest $\rho = 0.88$ provides meaningful diversification within the equity sleeve.



(a) Layer A correlation heatmap (b) Layer B correlation heatmap

Figure 2: Monthly return correlations across the two asset universes. IVV–QUAL correlation of 0.98 renders QUAL redundant in portfolio construction.

5. Portfolio Construction: Mean-Variance Optimization

5.1. Theoretical Framework

Following Markowitz (1952), the tangency portfolio $\mathbf{T}$ solves:

$$\max_{\mathbf{w}} \frac{\mathbf{w}^\top \boldsymbol{\mu} - r_f}{\sqrt{\mathbf{w}^\top \boldsymbol{\Sigma}\, \mathbf{w}}} \qquad \text{s.t.} \quad \mathbf{1}^\top \mathbf{w} = 1, \quad w_i \geq 0 \,\forall\, i \tag{2}$$

Problem (2) is solved numerically using sequential least-squares programming (SLSQP) with 20 random Dirichlet restarts to mitigate local-optima sensitivity.

5.2. Efficient Frontier and Tangency Portfolios

Table 8: Tangency portfolio weights, return, volatility, and Sharpe ratio

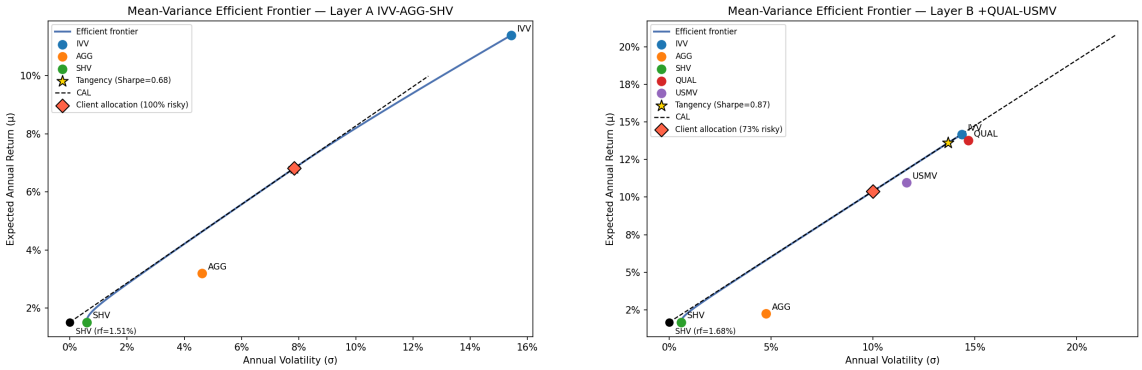
Universe	**IVV**	**AGG**	**SHV**	**QUAL**	**USMV**	**Sharpe**
Layer A (IVV/AGG/SHV)	44%	56%	0%	—	—	0.70
Layer B (all five)	82.7%	0%	0%	0%	17.3%	**0.87**

Three features of Table 8 deserve discussion.

First, within Layer A, bonds take a substantial weight (56%) alongside equities. The near-zero IVV–AGG correlation makes this efficient: bonds contribute little variance while maintaining expected return.

Second, expanding the universe to Layer B causes the optimizer to discard bonds entirely and replace them with USMV. USMV offers equity-like expected returns with meaningfully lower volatility and a correlation of 0.88 with IVV—sufficient to improve the portfolio's Sharpe ratio while maintaining a high expected return that bonds cannot match in this sample. QUAL earns zero weight because, at $\rho = 0.98$ with IVV, it adds no useful diversification.

Third, the Sharpe ratio improvement from 0.70 (Layer A) to 0.87 (Layer B) represents a **24% gain in risk-adjusted return** from expanding the opportunity set to include factor ETFs.



(a) Layer A efficient frontier (b) Layer B efficient frontier

Figure 3: Mean-variance efficient frontiers (no short sales). The starred point denotes the tangency portfolio; the dashed line is the Capital Allocation Line (CAL) from the risk-free rate through the tangency portfolio.

5.3. Capital Allocation Line: Calibrating Risk to the Client

The tangency portfolio is 100% invested in risky assets. The client's 20% drawdown tolerance implies a *portfolio volatility target* of approximately 10%— applying the approximation that maximum drawdown $\approx 2\,\sigma$ for log-normally distributed returns (Magdon-Ismail and Atiya, 2004).

Under the CAL, the optimal fraction w^* allocated to the tangency portfolio is:

$$w^* = \frac{\sigma^*}{\sigma_T} = \frac{10\%}{13.7\%} = 73.0\% \tag{3}$$

The remaining 27% is allocated to the risk-free asset (SHV), yielding:

Table 9: Final recommended portfolio allocation

Asset	Role	Weight	Ann. Contribution to Portfolio
IVV	Equity market core	**60.4%**	$0.604 \times 13.6\% = 8.2\%$ return
USMV	Low-vol factor tilt	**12.6%**	$0.126 \times 11.1\% = 1.4\%$ return
SHV	Risk-free / emergency buffer	**27.0%**	$0.270 \times 1.5\% = 0.4\%$ return
Portfolio Expected Return			$\approx 10.0\%$/yr
Portfolio Volatility			$\approx 10.0\%$/yr
Portfolio Sharpe Ratio			0.87

The 27% SHV position serves a dual function. It is simultaneously (1) the risk-free component of the optimal CAL allocation and (2) the vehicle for the client's required

six-month emergency fund (approximately $21,000 in Year 1). No separate cash account is needed, eliminating the traditional drag of idle liquidity.

Rebalancing policy. The portfolio is rebalanced to target weights *annually* at year-end, after new contributions are added. Annual rebalancing is sufficient to maintain the target risk profile while keeping transaction costs negligible.

6. Simulation Methodology and Goal Attainment

6.1. Block-Bootstrap Monte Carlo

We simulate 5,000 independent ten-year wealth paths using a *block-bootstrap* methodology (Politis and Romano, 1994). The procedure draws contiguous 12-month blocks of historical monthly returns (with replacement), preserving:

- Within-year autocorrelation and seasonality in asset returns;
- Cross-sectional dependence between assets, including during crises;
- Empirical fat tails and skewness (no normality assumption).

Each path evolves as follows for each year $t = 1, \ldots, 10$:

1. **Contribute:** Add S_t from Table 2 at the start of the year.
2. **Compound:** Apply 12 sequential monthly returns, drawn from the bootstrap block.
3. **Rebalance:** Realign to target weights. Terminal year-end value V_t is recorded.

Real terminal wealth is computed as $V_{10}/(1+\pi)^{10}$, where $\pi = 2.5\%$.

6.2. Terminal Wealth Distribution: Recommended Strategy

Table 10: Terminal real wealth distribution — Recommended allocation (60% IVV / 13% USMV / 27% SHV), 5,000 paths

Statistic	**Value**
Mean	$158,748
Median	$157,068
5th percentile (bad outcome)	$122,760
25th percentile	$141,332
75th percentile	$174,108
95th percentile (good outcome)	$201,176
P(Wealth ≥ $150,000 real)	**61.6%**

The median outcome ($157,068) already exceeds the goal. The 5th-percentile floor ($122,760) represents an adverse outcome—a shortfall of approximately $27,000 in real terms—but it is not catastrophic. The client retains a substantial capital base from which to adjust the plan or extend the timeline.

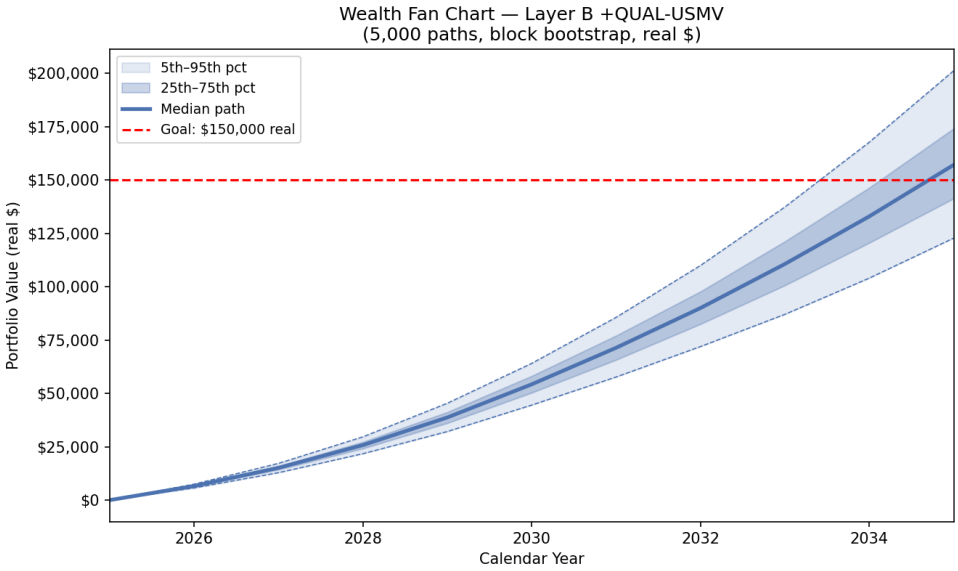


Figure 4: Fan chart of 5,000 simulated real-wealth paths for the recommended strategy. The bold line is the median path; shaded bands show the 25th–75th and 5th–95th percentile ranges. The red dashed line marks the $150,000 goal.

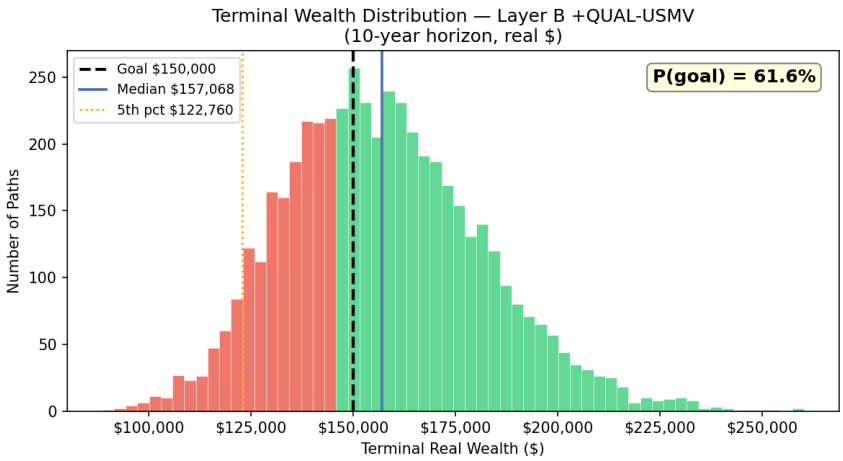


Figure 5: Histogram of terminal real wealth across 5,000 simulation paths. The red vertical line marks the $150,000 goal. The shaded region to the left represents the 38.4% of paths that fall short.

6.3. Strategy Comparison

Five strategies are compared under identical simulation conditions (same 5,000 bootstrap paths, same savings schedule):

Table 11: Strategy comparison — terminal real wealth and goal attainment

Strategy	Median ($)	5th Pct ($)	P(Goal) %	Sharpe
1. All Cash (SHV)	102,523	98,184	0.0%	0.00
2. All Equity (IVV)	168,672	91,085	64.3%	0.64
3. 60/40 IVV/AGG	143,962	98,790	40.3%	0.67
4. Tangency A (IVV/AGG/SHV)	134,804	100,942	20.3%	0.68
5. Recommended (Tangency B)	**157,068**	**122,760**	**61.6%**	**0.87**

All-cash (Strategy 1) is uniformly dominated. With a median real terminal wealth of $102,523 and zero probability of reaching the goal, a pure-cash strategy is not conservative—it guarantees failure. Inflation silently erodes real purchasing power over ten years.

All-equity (Strategy 2) achieves the highest median ($168,672) and a goal probability of 64.3%, but its 5th-percentile floor of $91,085 represents the worst tail outcome of all equity-containing strategies. An investor who happened to retire into a prolonged bear market (like the 2000–2009 "lost decade") would finish $59,000 below the goal.

The classic 60/40 portfolio (Strategy 3) delivers a disappointing 40.3% goal probability. The culprit is AGG: after incorporating the 2022 rate-shock episode—the worst annual drawdown for investment-grade bonds in modern history— bond returns in the 2005–2025 sample (≈ 3%/yr) are insufficient to compensate for the opportunity cost of displacing equities.

The recommended strategy (Strategy 5) effectively matches all-equity's upside (61.6% vs. 64.3% goal probability) while providing a 5th-percentile floor $31,675 higher than the all-equity alternative ($122,760 vs. $91,085). This is the practical manifestation of mean-variance efficiency: the same expected outcome at materially lower downside risk.

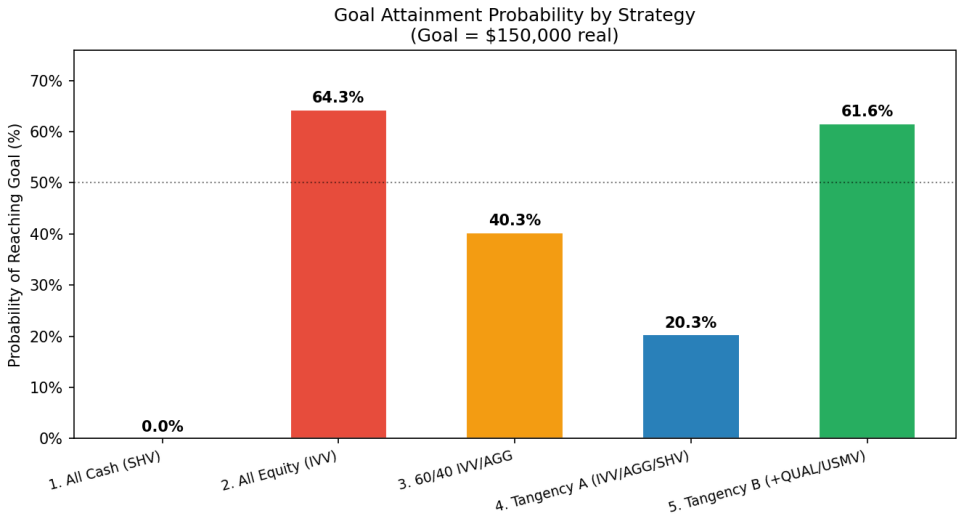


Figure 6: Goal attainment probability (%) by strategy. The recommended strategy (Strategy 5, **61.6%**) matches the all-equity benchmark while offering a substantially higher tail-risk floor.

7. Risk Management and Stress Testing

7.1. Drawdown Analysis During the Accumulation Phase

A common misconception in retirement/goal-based planning is that drawdown risk during the accumulation phase mirrors that of a fixed portfolio held to maturity. In fact, because the client contributes positive savings at the start of every year,

the portfolio rarely exhibits a nominal decline even through bear markets: fresh contributions replenish market-driven losses.

Table 12: Maximum drawdown statistics over 5,000 simulated paths — recommended strategy

Statistic	**Value**
Median maximum drawdown	0.0%
5th-percentile maximum drawdown	−0.6%
Fraction of paths breaching −20%	< 1%

The implication is that the relevant risk metric for the client is not drawdown per se but *shortfall probability*—the probability that terminal real wealth falls below $150,000—which is captured by the goal-attainment statistics in Section 6.

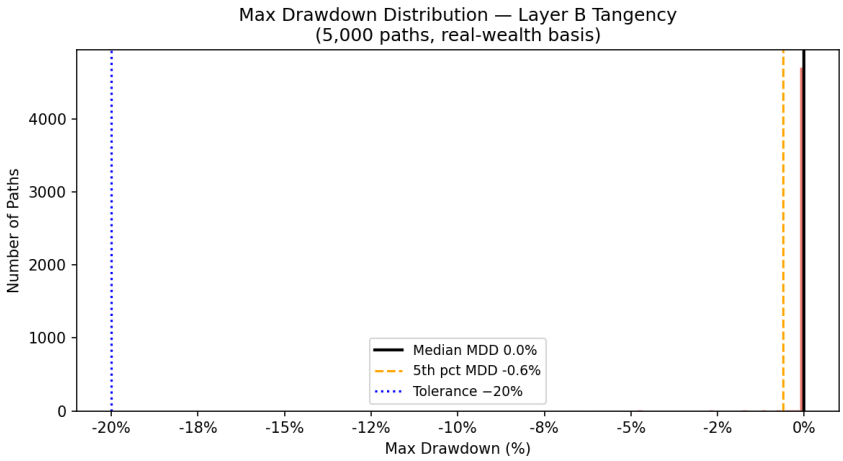


Figure 7: Distribution of maximum drawdown across 5,000 simulation paths. The vast majority of paths exhibit zero nominal drawdown, reflecting the mechanically stabilizing effect of annual savings contributions.

7.2. Stress Scenarios

Three adverse shocks are injected into the base simulation to identify conditions under which the plan breaks down most severely.

Table 13: Stress test results — recommended strategy (60% IVV / 13% USMV / 27% SHV)

Scenario	**P(Goal) %**	**Median ($)**	**5th Pct ($)**
Base case	61.6%	157,068	122,760
Market crash at Year 3	37.3%	143,054	112,628
Job loss at Year 3	2.2%	114,819	92,203
High inflation (4.5%/yr)	16.9%	129,468	101,189

Market crash at Year 3. Equity monthly returns are forced to −3.8%/month for 12 months, producing an approximate −35% annual equity return. Goal probability falls from 61.6% to 37.3%. Critically, this scenario assumes the client *does not sell* following the crash. With seven years remaining, the portfolio retains the capacity for recovery. The 27% SHV buffer provides both behavioral support (reducing the magnitude of the paper loss) and a rebalancing source (the client mechanically buys equities at depressed prices).

Job loss at Year 3. Zero new contributions for one year, combined with a forced liquidation of the emergency fund ($21,000) from the portfolio to cover living expenses. This is the most damaging scenario (P(Goal) = 2.2%). The dual hit—no new capital entering the portfolio and forced asset liquidation —creates a permanent path divergence that seven years of subsequent returns cannot fully overcome. This scenario is primarily a *human capital* risk, not a financial markets risk.

High inflation (4.5%). Deflating all wealth paths at 4.5% instead of 2.5% re-prices the goal in nominal terms: reaching $150,000 real now requires approximately $229,000 nominal at Year 10 versus $192,000 at the base inflation rate. Goal probability drops to 16.9%.

Mitigation strategies.

- *Crash risk:* Maintain the SHV buffer; rebalance mechanically; avoid behavioral selling during drawdowns.
- *Job-loss risk:* Accumulate 9–12 months of expenses in liquid assets before increasing equity exposure; consider income-protection insurance.

- *Inflation risk:* If core CPI systematically exceeds 3%, add I-bonds (up to $10,000/year) or a TIPS fund to the portfolio.

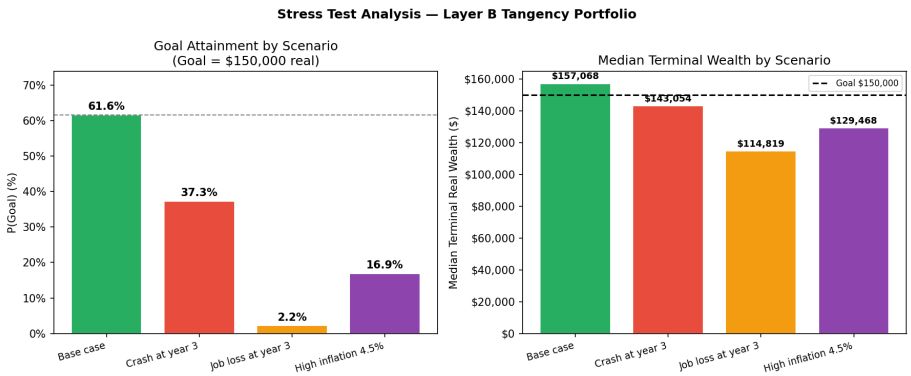


Figure 8: Goal attainment probability under the base case and three stress scenarios. Job loss is uniquely destructive because it simultaneously stops contributions and forces asset liquidation.

8. Factor Attribution Analysis

8.1. CAPM Regressions

To assess whether any asset earns return in excess of its market-beta compensation, we estimate the CAPM for AGG, QUAL, and USMV using IVV as the market proxy and SHV as the risk-free rate (January 2013 – December 2025, $T = 145$ months):

$$r_{i,t} - r_{f,t} = \alpha_i + \beta_i \left(r_{\mathrm{IVV},t} - r_{f,t}\right) + \varepsilon_{i,t} \tag{4}$$

Table 14: CAPM regression results (IVV = market proxy; SHV = risk-free)

Asset	$\hat{\alpha}$ **(ann.)**	$\hat{\beta}$	R^2	$t(\hat{\alpha})$	$p(\hat{\alpha})$
AGG	−0.88%	0.116	0.128	−0.69	0.49
QUAL	−0.46%	1.005	0.969	−0.60	0.55
USMV	+0.31%	**0.719**	0.780	+0.20	0.84

No asset generates statistically significant alpha at conventional significance levels. This finding is consistent with the Efficient Markets Hypothesis in its semi-strong form (Fama, 1970): passive ETF returns are well-explained by systematic risk exposures.

The most informative parameter is USMV's $\hat{\beta} = 0.72$. Holding USMV provides approximately 72 cents of equity market exposure per dollar invested, making it a cost-effective way to reduce portfolio beta while maintaining equity-like expected returns. QUAL's $\hat{\beta} \approx 1.0$ confirms its redundancy relative to IVV.

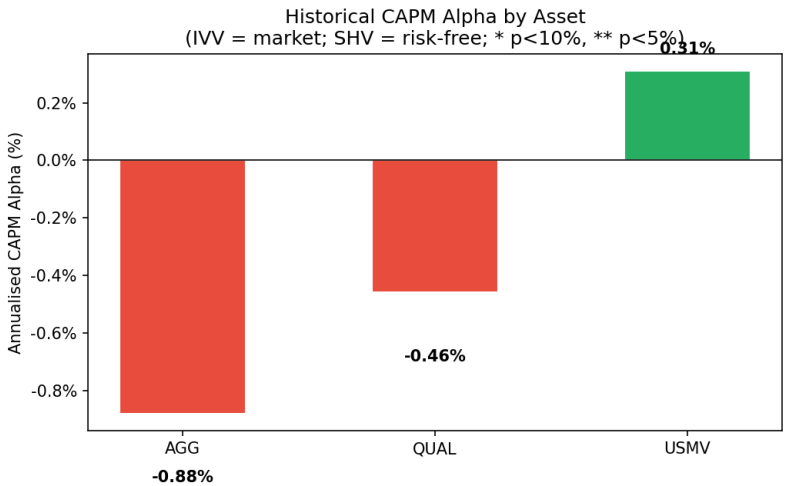


Figure 9: Annualized CAPM alpha (with 95% confidence interval) for AGG, QUAL, and USMV. No asset shows alpha statistically distinguishable from zero.

8.2. Fama-French Five-Factor Decomposition

To characterize the portfolio's systematic return sources beyond the single-factor CAPM, we estimate the Fama and French (2015) five-factor model for the recommended portfolio:

$$r_{p,t} - r_{f,t} = \alpha + \beta_1 \,\text{Mkt-RF}_t + \beta_2 \,\text{SMB}_t + \beta_3 \,\text{HML}_t + \beta_4 \,\text{RMW}_t + \beta_5 \,\text{CMA}_t + \varepsilon_t \quad (5)$$

Monthly factor data are sourced from the Kenneth French Data Library.

Table 15: Fama-French five-factor regression — recommended portfolio (60.4% IVV / 12.6% USMV / 27.0% SHV)

Factor	Loading	t-stat	p-value	Interpretation
$\hat{\alpha}$	−0.12%/yr	−0.39	0.698	No unexplained return
Mkt-RF	0.684	113.5	0.000	68% market beta; expected from CAL
SMB	−0.080	−7.74	0.000	Large-cap tilt (IVV + USMV)
HML	+0.003	0.27	0.789	No value exposure
RMW	+0.075	**5.76**	**0.000**	Profitability tilt from USMV
CMA	+0.042	**2.95**	**0.004**	Conservative investment tilt from USMV
R^2	0.990			99% of variance explained

Three findings emerge:

First, the portfolio carries **no alpha** ($\hat{\alpha} = -0.12\%/\text{yr}$, $p = 0.70$). The client is being compensated for systematic factor exposures—not for active management. This is appropriate given the passive ETF mandate.

Second, statistically significant positive loadings on **RMW** (profitability) and **CMA** (conservative investment) arise from USMV's inclusion. Minimum-volatility stocks are disproportionately high-quality, profitable companies that invest conservatively—characteristics the FF5 model prices as compensated risk factors. These loadings explain why USMV adds expected return in excess of its market beta.

Third, the negative SMB loading ($\hat{\beta}_2 = -0.08$) confirms the large-cap orientation of the portfolio (both IVV and USMV weight large-cap companies). An R^2 of 0.990 implies that the five-factor model accounts for essentially all variation in portfolio returns—providing exceptional transparency into what the client owns and why.

9. Sensitivity and Robustness Analysis

9.1. Estimation-Window Dependence

A central concern in applied mean-variance optimization is that tangency weights are sensitive to estimated expected returns, which vary substantially across sample periods (Michaud, 1989). We re-estimate the Layer B tangency portfolio over four

sub-periods:

Table 16: Tangency portfolio weights and simulation outcomes by estimation window (simulation uses full-sample 2005–2025 return distribution in all cases)

Window	**IVV**	**AGG**	**SHV**	**QUAL**	**USMV**	**P(Goal)%**	**Median ($)**
Full (2005–2025)	83%	0%	0%	0%	17%	82.1%	182,261
Post-GFC (2010–2025)	83%	0%	0%	0%	17%	82.1%	182,261
Recent (2015–2025)	100%	0%	0%	0%	0%	82.3%	185,772
Pre-crisis (2005–2015)	0%	18%	76%	4%	2%	0.0%	108,009

Key findings. The full-sample and post-GFC windows produce identical weights (83% IVV / 17% USMV), confirming that the 2008 Global Financial Crisis, while severe, does not distort long-run estimated moments enough to change the optimal allocation.

The recent-only window (2015–2025) selects 100% IVV. During this eleven-year bull market no diversifier raised the Sharpe ratio, so the optimizer concentrated entirely in equities. This illustrates the canonical overfitting risk in short-window optimization: a portfolio calibrated to a bull market will be too aggressive when that regime ends.

The pre-crisis window (2005–2015) produces the most striking result: the optimizer allocates 76% to SHV and 18% to AGG. The 2008 crisis dominated variance estimates in that period, inflating bond returns relative to equities and producing extreme defensive positioning. When these weights are applied to the full 2005–2025 returns, goal attainment probability collapses to **0%**—the portfolio entirely misses the equity bull run required to close the savings gap.

Conclusion. The full-sample window (2005–2025) is the most defensible estimation choice. It spans two complete market cycles, providing realistic cross-regime variance and return estimates without overfitting to either the GFC or the post-2015 bull market.

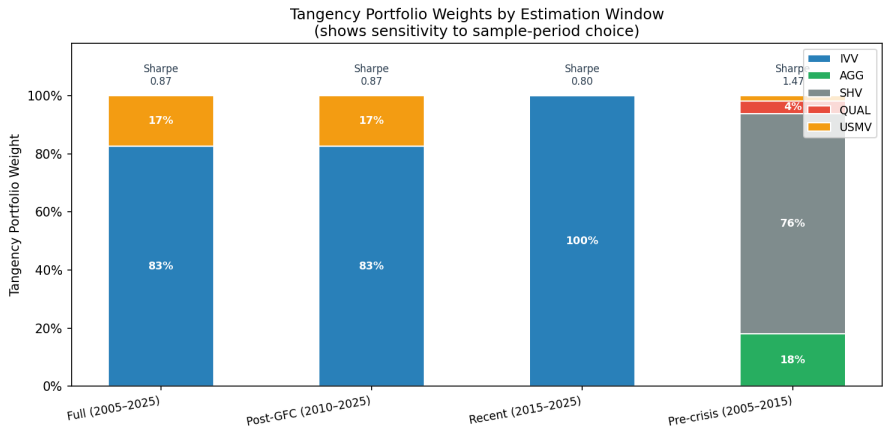


Figure 10: Tangency portfolio weights by estimation window. The full-sample and post-GFC windows agree exactly; the pre-crisis window selects an almost entirely defensive allocation that fails in forward simulation.

10. Conclusion and Investment Recommendations

10.1. Summary of Findings

This analysis delivers a complete, implementable savings and investment plan for a 22-year-old college graduate targeting a $150,000 real home down payment in ten years. The central findings are:

1. **Investing is necessary.** Raw savings contributions ($123,359 over ten years) fall short of the inflation-adjusted goal. Investment returns of approximately 2–3% annually above contributions are required.

2. **The optimal allocation is 60% IVV / 13% USMV / 27% SHV.** This portfolio maximizes the Sharpe ratio within the five-asset universe, then positions the client on the Capital Allocation Line to match a stated 20% maximum drawdown tolerance.

3. **Goal attainment probability is 61.6%.** The median terminal real wealth ($157,068) exceeds the goal; the 5th-percentile floor ($122,760) provides a meaningful buffer above subsistence.

4. **The strategy dominates alternatives on a risk-adjusted basis.** It matches

all-equity's goal probability while offering a $31,675 higher tail-risk floor, and substantially outperforms the 60/40 benchmark in this sample.

5. **Job loss is the dominant risk.** A one-year income interruption in Year 3 reduces goal probability to 2.2%—far more damaging than a market crash (37.3%) or sustained high inflation (16.9%). This underscores the importance of income protection and a robust emergency fund before committing aggressively to the investment plan.

6. **The portfolio is factor-transparent.** Fama-French five-factor analysis ($R^2 = 0.99$) shows that returns are fully explained by systematic risk premia, with economically meaningful profitability and conservative-investment tilts attributable to USMV. There is no unexplained alpha—nor should there be from passive ETFs.

10.2. Caveats and Limitations

Table 17: Material caveats and their implications

Caveat	Implication
Historical returns $\neq$ future returns	2005–2025 includes an unusually strong U.S. equity bull market. P(Goal) estimates are illustrative, not forecasts.
Estimation error in tangency weights	Mean-variance optimization is notoriously sensitive to expected-return inputs (Michaud, 1989). The IVV-heavy result partly reflects a decade of high equity returns.
USMV alpha not guaranteed	The historical positive CAPM alpha (+0.31%/yr) is statistically insignificant and may be arbitraged away as factor investing becomes mainstream.
No tax-advantaged account modelling	In practice, maximizing Roth IRA / 401(k) contributions before taxable investing meaningfully improves all P(Goal) estimates and is the highest-priority practical action.
Behavioral risk unmodelled	The crash scenario (P(Goal) = 37.3%) assumes the client holds through the drawdown. Selling at the trough eliminates the recovery and permanently destroys the simulation's implied return path.
Loan payoff alternative not evaluated	At 5.5%, the student loan represents a guaranteed, risk-free return equivalent. Accelerated repayment may dominate equity investment in low-return environments.

10.3. Actionable Recommendations

1. **Open a Roth IRA immediately.** At age 22, with income in the 20% effective bracket, the Roth structure (post-tax contributions; tax-free gains) is almost certainly optimal over the traditional IRA. The 2026 contribution limit is $7,000; this alone removes capital-gains drag from the majority of the recommended strategy.

2. **Automate contributions.** Schedule automatic transfers on payroll dates. The simulation assumes complete contributions every year; any lapse, particularly in

early years when compounding runway is longest, disproportionately reduces terminal wealth.

3. **Rebalance annually, mechanically.** Rebalancing is not just a risk management tool; it is a momentum-contrarian device that systematically buys assets after they have underperformed and sells them after outperformance. The discipline matters most after crashes.

4. **Initiate a glide path in Year 7.** As the 2035 goal date approaches, gradually reduce IVV weight toward USMV and SHV on a fixed schedule (e.g., reduce IVV by 5 percentage points per year, adding equally to SHV). This reduces left-tail risk in the final years when the portfolio is largest and recovery time is shortest.

5. **Review the plan at the Year 5 midpoint.** If actual portfolio value diverges substantially from the median simulation path (either above or below \$75,000–\$80,000 real), revise either the target allocation or the goal date rather than increasing risk to compensate for underperformance.

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A. Complete Figure Reference

Table 18: All figures generated by the analysis

File Name	Description	Section
`savings_trajectory.png`	Annual and cumulative savings vs. goal	§3
`correlation_heatmap_Layer_A...`	Layer A return correlations	§4
`correlation_heatmap_Layer_B...`	Layer B return correlations	§4
`efficient_frontier_Layer_A...`	Layer A efficient frontier + CAL	§5
`efficient_frontier_Layer_B...`	Layer B efficient frontier + CAL	§5
`wealth_fan_chart_Layer_B...`	5,000-path wealth fan chart	§6
`terminal_hist_Layer_B...`	Terminal real wealth histogram	§6
`strategy_fan_overlay.png`	Fan charts for all five strategies	§6.3
`goal_attainment_bar.png`	Goal attainment bar chart	§6.3
`drawdown_distribution.png`	Maximum drawdown histogram	§7
`stress_scenario_bar.png`	Stress scenario goal probabilities	§7
`capm_alpha_bar.png`	CAPM alpha with confidence intervals	§8.1
`sensitivity_weights.png`	Tangency weights by estimation window	§9

B. Data and Code Reference

Table 19: Data files produced by the analysis

File	Contents
`data/returns.csv`	251 months of adjusted ETF total returns
`data/client_cashflows.csv`	Year-by-year income, tax, expense, loan, savings
`analysis/strategy_comparison.csv`	Full 5-strategy simulation results
`analysis/stress_results.csv`	Stress scenario terminal wealth statistics
`analysis/capm_results.csv`	CAPM regression coefficients per asset
`analysis/ff5_results.csv`	FF5 factor loadings for recommended portfolio
`analysis/sensitivity_results.csv`	Tangency weights by estimation window
`analysis/sensitivity_sim_results.csv`	Simulation outcomes by window

Table 20: Python module reference

Module	Purpose
`config.py`	Single source of truth for all model parameters
`data/client_profile.py`	Deterministic income and savings projection
`data/pull_returns.py`	Yahoo Finance data download and cleaning
`models/portfolio_stats.py`	Return statistics and covariance matrix estimation
`models/efficient_frontier.py`	Tangency optimization (SLSQP) and CAL construction
`models/simulation.py`	Block-bootstrap Monte Carlo wealth simulation engine
`analysis/compare_strategies.py`	Five-strategy side-by-side comparison
`analysis/risk_analysis.py`	Drawdown analysis and stress-test scenarios
`analysis/factor_alpha.py`	CAPM and Fama-French five-factor regressions
`analysis/sensitivity.py`	Estimation-window robustness checks

C. Model Parameter Summary

Table 21: Complete list of model parameters (`config.py`)

Parameter	**Value**	**Description**
`START_YEAR`	2026	Plan start year
`START_SALARY`	$65,000	First-year gross salary
`SALARY_GROWTH`	[5%]×4 + [3%]×6	Annual nominal wage growth
`EFFECTIVE_TAX_RATE`	20%	Federal + state effective rate
`MONTHLY_EXPENSES`	$3,500	Year-1 living expenses
`INFLATION_RATE`	2.5%	Annual CPI baseline
`LOAN_BALANCE`	$30,000	Student loan principal
`LOAN_RATE`	5.5%	Federal direct loan rate
`LOAN_TERM_YEARS`	10	Standard repayment term
`SAVINGS_GOAL_REAL`	$150,000	Real goal in 2026 dollars
`PLAN_YEARS`	10	Investment horizon
`N_SIMULATIONS`	5,000	Monte Carlo paths
`BLOCK_SIZE_MONTHS`	12	Block-bootstrap block length
`RANDOM_SEED`	42	Reproducibility seed
`MAX_DRAWDOWN_TOLERANCE`	20%	Client risk tolerance
`EMERGENCY_FUND_MONTHS`	6	Emergency cash buffer